

Deep learning for Aspect-based Sentiment Analysis

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Abstract. With the popularity of the Internet and the amount of information people can get from the Internet increases exponentially, sentiment analysis depending on text information has become a rather important tool in daily life, which helps people improve the quality of products and their lifestyle. According to the different granularity of sentiment text, sentiment analysis can be divided into three levels: word, sentence, and document level. While coarse-grained sentiment analysis only focuses on the polarity of emotions and neglects the finer-grained tendency of emotions. Driven by this demand, some researchers put forward the aspect-based sentiment analysis. ABSA can be categorized into two basic steps: aspect extraction and sentiment classification, on which researchers are mainly focusing to improve its performance. Deep learning has gradually become a research hotspot in recent years and has been widely used in CV and NLP fields. Recently, some efforts have begun to apply deep learning to ABSA tasks. In this paper, we further summarize and analyze the recent achievements of ABSA in the application of different deep learning models in terms of their completeness, accuracy, and detail. We also summarize ABSA's current challenges and difficulties and look forward to the future development and improvement of ABSA based on the existing experience.

Keywords: sentiment analysis; aspect-based sentiment analysis; deep learning; convolution neural network; recurrent neural network

I. Introduction

With the popularity of online social media, more and more people begin to communicate via the Internet, which is a public platform with traits of transmission, freedom, openness, real-time, and sharing. Based on the above characteristics of the Internet, people interact with others through text messages on the Internet, such as comments about people, events and products. These text messages own great research value and are also on a large scale. Having provided with so much plenty of informative and real-time information, people have to use them and contribute to their daily lives.

Sentiment analysis (SA) in NLP (Natural Language Process), which is also called opinion mining or tendency analysis, has become a hot and active filed in recent years. Such User Generated Content (UGC) usually contains a large number of subjective emotions. Through sentiment analysis, people can get to know the user's emotional tendency towards a specific term. At present, SA has been applied to many fields, such as marketing, finance, medical care, and even history, and it all makes good progression. According to the different granularity of sentiment text, SA can be roughly divided into

three levels: word, sentence, and document. Sentence and document level SA research generally only summarize one emotional tendency (positive, negative, neutral, etc.) analyzing one paragraph (or one sentence). At the same time, the information obtained in the text usually contains a lot of information inside, and maybe only one sentence will give it different evaluations for different entities. To better use the sentimental text information people get, they usually choose to use ABSA (Aspect-Based Sentiment Analysis) to carry out fine-grained sentiment analysis to mine the maximum source information of the text. For example, this text can be analyzed in different level of SA with a different result: "This PC has a sharp screen, appendix with flexible mouse and keyboard, while a bit sluggish playing games", with sentence or text level analysis, people can get a neutral or much positive assessment, but with ABSA, a much more detailed analysis of different parts of the PC is achieved, including clarity, fluency, accessories, and several aspects. This fine-grained analysis provides a more granular reference for the vendor to future improvements on products.

ABSA task typically contains two sub-tasks: aspect extraction and sentiment classification. In the process of aspect extraction, the extracted aspect can be either an entity that appears explicitly in a sentence (Target, Aspect Term) or an aspect category that describes things implicitly. Based on this, two sub-tasks of aspect extraction--Aspect Category Sentiment Analysis (ACSA) and Aspect Term Sentiment Analysis (ATSA)--are extended. These two sub-tasks face different entities in the text, and their SA process will make different improvements and applications according to their characteristics. This paper also introduce different methods and understandings of the two sub-tasks in this paper. In the sentiment classification stage, deep learning neural network improves and optimizes sentiment classification by using the extracted features to carry out the word-level label or semantic analysis of the whole sentences (using language rules, emotion words, negative words, and degree adverbs). The ABSA method analyzed in this paper is mainly aimed at the above two aspects to make optimization and improvement.

Traditional methods for ABSA include statistical methods [1] and machine learning methods [2], but there are several problems. ABSA, based on statistical methods, mainly depends on an emotion dictionary, which classifies words into general emotional words, degree adverbs, negative words, and domain words. It matches words in the text with an emotional

dictionary and mine positive and negative information. This method is simple, generalized, and easy to operate, but it owns several flaws such as low precision, difficulty of constructing the emotional dictionary and needs of a continuous update. The performance of it mainly depends on the emotional dictionary. Moreover, in the application of neural network model, which requires a lot of evaluation in the field of specific text information input (this information is in a variety of forms, or it may be difficult to find a large scale of data in a unified form), also need to deal with the characteristics of the label for text information (manual processing), is a huge task. In this case, many researchers using different deep learning approaches based on CNN/RNN and put forward many optimized solutions.

The research of ABSA has changed and promoted every year and make innovation in various aspects. In order to better summarize the recent development of ABSA, a comprehensive analysis and investigation report is done in this paper. This report is not a restatement of existing research, but to summarize and compare the innovation and development of deep learning methods in the SA field, reflect on the existing problems, and put forward some new suggestions to contribute to ABSA's future research. Reading this review can provide novices or researchers devoting on ABSA's research paper with a reference that can help them make a quick understanding of current work and developments of deep learning techniques in the ABSA task.

The remainder of this paper is as follows: Section 2 defines ABSA tasks in more detailed and introduces datasets. Section 3 analyzes ABSA's deep learning methods, which are divided into two modes based on RNN and CNN and the problems and challenges targeted and solved by each deep learning method. Section 4 mainly introduces the difficulties and challenges that ABSA and Sentiment Analysis still face.

II. Aspect-Based Sentiment Analysis (ABSA)

A. ABSA and two sub-tasks

With the spread of the Internet and the development of social software, more and more people air their opinions online, just like the comments about products, people, or news. With the large-scaled data from social media, sentiment analysis becomes a crucial task in NLP.

Sentiment analysis, also known as opinion extraction, opinion mining, sentiment mining, and subjectivity analysis. It is the process of analyzing, processing, inducting, and reasoning the subjective text with sentiments or emotions. SA contains three types of tasks: simple task, more complex task, and advanced task, three tasks reflect the complexity of SA. With the different tasks, SA is used in different environments to solve various problems. For instance, the simple task only judges two poles of emotion, "positive" or "negative", which can be used to solve movie classification problems. A more complex task does rank the attitude, which can be applied in evaluating products. The advanced task can detect more complex attitude types, which can be used to forecast stock markets. Recently, SA has grown into one of the most active

research areas in natural language processing (NLP). In fact, SA has spread from the computer since to management and social sciences such as marketing, finance, political science, communications, medical science, and even history. As can be seen, SA has permeated every aspect of our lives. With the basis of extensive learning from previous research, sum up experience, constantly ask questions, solve problems and gradually grow this field.

Previous work only focuses on obtaining sentiments or opinions at the sentence or document level, but many entities have different attributes in reviews. For achieving the goal, the aspect-based sentiment analysis (ABSA) task is proposed. ABSA is the task of identifying fine-grained opinion polarity towards a specific aspect associated with a given target, which is used to process information.

And there are two sub-tasks about ABSA: aspect extraction and sentiment classification. Aspect extraction refers to the extraction of specific attributes of things from comments and corresponding sentiment words, which is the basis for ABSA. Sentiment classification refers to identifying the sentiment polarity of the extracted evaluation features, which is the key to ABSA. For example, from this sentence "This book has beautiful illustrations, but the content is rather dull.", we can find two aspects: illustrations and content in aspect extraction. Then sentiment polarity (SP) can be speculated to "positive" or "negative" for every sentiment of the target aspect. In the above sentence, one aspect is "illustrations", and the SP is "positive" because of "beauty". The other aspect is "content", and the SP is "negative" because of "rather full". Based on this example, two tasks of ABSA can be understood well.

B. Datasets of ABSA

Now ABSA is widely used in evaluating reviews from most social media, which includes comments on products from Amazon or Taobao and comments on topics from Twitter or Instagram. When dealing with a specific topic, multiple aspects of it may be evaluated. It is useful to use ABSA to further analysis and explore the potential value of data.

In recent years, different kinds of datasets are used. The dataset from Hu et al. (2004) [3] is frequently used to mine and summarize customer reviews about electronic products. Further, some data on restaurants and laptops are released by 'International Workshop on Semantic Evaluation' (SemEval 2014-2016) [4]. Some datasets about movie review are developed, such as IMDB [5], which contains 50,000 comments, of which 25,000 are used as a training set and the rest as a test set, and the label is "positive" and "negative". Business, user, review, tips, and check in data can also be got from the Yelp dataset [6].

Besides, more and more Chinese datasets are being generated and applied to ABSA. ThucNews dataset [7] is generated by filtering the historical data of Sina News RSS subscription channel from 2005 to 2011, which contains 14 candidate categories, like finance, lottery, property, education, etc. The Toutiao news text classification dataset has 382,688

entries data size, distributed in 15 classifications. SogouCA and SougoCS datasets [8] owns news data from 18 different channels providing URLs and text information. Besides, some hotel review data and reviews about food delivery can also be found in the ChnSentiCorp_hlt_all dataset [9] and waimai_10k dataset, respectively.

C. Previous approaches to ABSA tasks

As mentioned above, ABSA has two tasks, aspect extraction and sentiment classification, which are the foundation and key to ABSA, respectively. Aspect extraction mainly includes unsupervised methods based on frequency and pattern and supervised methods based on conditional random field (CRF) and deep learning. Among them, the earliest unsupervised method was proposed by Hu et al. (2004) [3], and they used association rule mining to extract attributes. Jakob et al. (2010) [10] used a Conditional Random Fields (CRF) model to extract evaluation features from film review data by selecting various features such as words, lexicality, and dependent syntax. Liu et al. (2018) [11] used recurrent neural network models (RNN) for the extraction of evaluative sentiment features. Poria et al. (2016) [12] used a convolutional neural network (CNN) approach to do automatic extraction of text features. Zeng et al. (2019) [13] proposed an aspect-level sentiment classification model based on a dual memory based on attention mechanism (BMAM) in a GRU-based decoder to achieve Fine-grained feature extraction.

Sentiment classification consists of unsupervised methods based on sentiment dictionaries and supervised methods based on machine learning and deep learning. The earliest supervised approach to sentiment classification was proposed by B. Pang et al. (2002) [14]. They used three supervised machine learning approaches: Naive Bayesian, support vector machine, and maximum entropy. Results state that the Naive Bayes approach outperformed the SVM. Han et al. (2020) [15] used a hierarchical neural network model based on the multi-attention mechanism to achieve word level, sentence level, and document level sentiment classification, further demonstrating the benefits of deep learning in sentiment analysis.

D. Evaluation measures of ABSA tasks

Once the modelling work has been completed, experimenters need to evaluate the model's effectiveness to check the validity of different approaches. To evaluate the various ABSA models well, four main measurements are proposed: Accuracy (Acc), Precision (P), Recall (R), and F-score (F1).

Before introducing the four evaluation indicators, the confusion matrix needs to be understood: True positive (TP), True negative (TN), False positive (FP), and False negative (FN). TP is a positive class that is predicted to be a positive class. TN is the negative class which is predicted to be a negative class. FP is the negative class that is predicted incorrectly to be positive. FN is a positive class that is predicted incorrectly to be a negative class. TP and FN are possible errors in the experimental results.

Accuracy is the most common evaluation metric: the number of samples that are scored correctly divided by the number of all samples. The formula is shown in equation (1) below, and in general, the higher the correct rate, the better the model. Precision is defined as shown in equation (2) below and represents the proportion of actual positive classes classified into positive classes. The recall is a coverage metric that counts multiple positive classes classified as positive, with the formula shown in equation (3) below, and in some cases, recall and sensitivity are the same. Sometimes Precision and Recall may have contradictions and then need comprehensive consideration. The most common method is to use the F-score, which is a weighted summed average of P and R. The formula is shown in equation (4) below, when the parameters $\alpha = 1$, F-score becomes F1, the formula is shown in equation (5) below, and when F1 is high it indicates that the experimental method is more effective.

$$Acc = (TP + TN) / (TP + TN + FP + FN) \quad (1)$$

$$P = TP / (TP + FP) \quad (2)$$

$$R = TP / (TP + FN) \quad (3)$$

$$F = ((\alpha^2 + 1)P \times R) / (\alpha^2(P + R)) \quad (4)$$

$$F1 = (2 \times P \times R) / (P + R) \quad (5)$$

III. Deep learning for Aspect-based Sentiment Analysis

Deep learning (DL) is a machine learning method that uses algorithms to learn data representations that take the outputs of one layer as inputs to the next layer, based on multi-layer modules that analyze and classify the inputs.

A. RNN-based ABSA

1) *Architectures.* A recurrent neural network (RNN) is able to handle a variable-length sequence input. The RNN handles the variable-length sequence by having a recurrent hidden state whose activation depends on that of the previous time. At any given time t , the current input is a combination of input at x_t and x_{t-1} . The output at any given time is fetched back to the network to improve on the output. The hidden state h_t is calculated as: acceptable:

$$h_t = s(W_{hh}h_{t-1} + W_{xh}x_{t-1}) \quad (6)$$

where W being the weighted matrix function between x_t and $h_{(t-1)}$, and s is a non-linear activation function, such as tanh or ReLU. Therefore, the output can be computed as:

$$y_t = W_h(y - 1)h_t \quad (7)$$

The LSTM can remove or add information to the cell state, carefully regulated by structures called gates. f is forget gate that decides which information to delete that is not important from previous time step. The sigmoid function looks at the previous state h_{t-1} along with the current input x_t and computes the function.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (8)$$

i_t is input gate that determines which information to let through based on its significance in the current time step. The sigmoid function decides which values to let through (0 or 1) and the tanh function give weight age to the values which are passed, deciding their level of importance (-1 to 1).

$$i_t = \sigma(w_i \cdot [h_{t-1}, x_t] + b_i) \quad (9)$$

$$\tilde{c}_t = \tanh(w_c \cdot [h_{t-1}, x_t] + b_c) \quad (10)$$

o_t is output gate that allows the passed in information to impact the output in the current time step. The sigmoid layer function decides what parts of the cell state make it to the output. Then, the cell state is put through tanh function to push the values to be between -1 and 1 and multiply it by the output of the sigmoid gate.

$$O_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (11)$$

$$h_t = o_t \cdot \tanh(C_t) \quad (12)$$

The principle behind GRU is very similar to LSTM which uses a gating mechanism to control input, memory and other information and make predictions at the current time step. The GRU has two gates, a reset gate and an update gate.

Intuitively, the update gate decides the amount of previous memories to be saved to the current time step. When x_t is plugged into the network unit, it is multiplied by its own weight W_z . The same goes for h_{t-1} which holds the information for the previous t-1 units and is multiplied by its own weight U_z . The update gate sums these two pieces of information and puts them into the Sigmoid activation function, thus compressing the activation result to between 0 and 1.

$$z_t = \sigma(W^z x_t + U^z h_{t-1}) \quad (13)$$

The reset gate determines how new input information is combined with previous memories. h_{t-1} and x_t are input and multiplied with their corresponding weights, the results are summed and put into a sigmoid function.

$$r_t = \sigma(W^r x_t + U^r h_{t-1}) \quad (14)$$

The current memory content is calculated by using the reset gate to store the relevant information from the past. Multiply the input x_t with a weight W and h_{t-1} with a weight U . Calculate the Hadamard (element-wise) product between the reset gate r_t and $U h_{t-1}$. That determines what to remove from the previous time steps. The nonlinear activation function tanh is used to produce the result of summing up the two steps.

$$h'_t = \tanh(W x_t + r_t \cdot U h_{t-1}) \quad (15)$$

$$h_t = z_t \cdot h_{t-1} + (1 - z_t) \cdot h'_t \quad (16)$$

2) Applications. To solve the limitation that CNN, LSTM and RNN can only capture intra-sentence relations but cannot capture inter-sentence relations. The text needs to be understood more comprehensive. This paper introduces a new method called H-LSTM (Hierarchy bidirectional LSTM). Hierarchy means it can make sentiment analysis in sentence level and review level, while bidirectional means it has forward and backward propagation to evaluate parameters. This model does not rely on any hand-engineered features or sentiment lexica. This model shows excellent performance in low-resource language with only a few hand-engineered features.

In order to identify the sentiment of an aspect of an entity, this paper proposes a new method called PhraseRNN for ABSA, which is an extension of RNN and AdaRNN. The basic idea is to make the representation of the target aspect richer by using syntactic information from both the dependency and constituent trees of a sentence. A new hierarchy is also constructed by integrating dependencies and phrases. The comparative experiment shows that their PhraseRNN outperformed other recent baselines.

To improve the current simple interaction mechanism and address the problem, it is unfair to use an average vector of the target to computing the attention score for context. A coattention mechanism was proposed which models both target-level and context-level attention alternatively. And then, they implemented the Coattention-LSTM network, Coattention-MemNet network, and a new location weighted function. They used SemEval 2014 Task 4 and Twitter to conduct the experiments and compared the methods with conventional memory networks to demonstrate the effectiveness.

B. CNN-based ABSA

1) Architectures. The motivation for using the CNN model in the ABSA task is to assume that keywords may contain aspect words and represent categories or determine polarity, regardless of their position. CNN was able to find these features through its architectural learning and extract local patterns from the data regardless of location. The CNN model has been applied to all tasks of ABSA, mainly in the consumer review domain. The basic single layer CNN for sentence modeling may consist of 4 layers. Input layer: representing the sentence of length n as:

$$x_{1:n} = x_1 * x_2 * \dots * x_n \quad (17)$$

where $x_i \in \mathbb{R}^d$ be the d-dimensional word vector corresponding to the i_{th} word in the sentence, and $*$ is the concatenation operator.

Convolutional layer: This layer is the first layer used to extract various features from the input sequence, where a mathematical operation of convolution is performed between the input image and a filter of a specific size.

$$s(i, j) = (X \cdot W)(i, j) + b = \sum_{k=1}^{n_{in}} (X_k \cdot W_k)(i, j) + b \quad (18)$$

where n_{in} is the number of input matrices or the dimension of the last dimension of the tensor and X_k is the k_{th} input matrix. W_k is the matrix of the sub-convolution kernel of the convolution kernel and $s(i, j)$ is the value of the corresponding position element of the output matrix corresponding to the convolution kernel W .

Max-pooling layer: The Pooling Layer usually serves as a bridge between the Convolutional Layer and the FC Layer. The primary aim of this layer is to decrease the size of the convolved feature map to reduce the computational costs. This layer selects the maximum value $\hat{c} = \max\{c\}$ as the feature corresponding to one particular filter.

Full-connected Layer: The Fully Connected (FC) layer consists of the weights and biases and the neurons that have a complete connection to all the activations from the previous layers. Their activation can be calculated using a matrix multiplication followed by a bias offset.

2) *Applications*. To solve the problem, most LSTM & attention mechanism is complicated and time-consuming on model training. The author proposes a new method based on CNN with a gating mechanism, which can output sentiment features according to the give aspect or entity via a Gated Tanh-ReLU Unit. And the cost of training time is reduced by the parallel computations of CNN. This model performs well on solving ACSA (context embedding), and ATSA (target embedding) 's limitations.

IV. Challenges

It is clear that the application of DL methods in ABSA is still immature. There are cases where the performance of DL methods is not as effective as expected. A major challenge for ABSA is the issue of domain adaptation, that is, whether parameters trained in one domain can be applied to another domain. Most noticeably, the performance of models varies significantly between domains. And ABSA has certain challenges in a multilingual environment, including word sense ambiguity, language-specific structure, and translation errors. Another challenge is imbalanced data. There are insufficient resources for many languages to construct NLP models. This is particularly the case for low-resource languages, which lack large monolingual or parallel corpora.

V. Conclusion

This review has explored an important research domain in NLP, i.e., Aspect-based Sentiment Analysis, a subfield of sentiment analysis that has accomplished many excellent researches. The survey has stated a thorough overview of the recent progress in ABSA by depicting the state-of-the-art deep-learning techniques. A model-based categorization of the recent solutions is presented, contributing to improving the process of aspect extraction, aspect sentiment analysis, or sentiment evolution. Key challenges are included as future research directions, which should be considered to achieve excellence in sentiment classification at the aspect level. Our survey can act as a foundation for researchers who want to know the recent progress of the field and help them formulate general strategies which would apply to most of the scenarios.

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